

AudSpeX

Global Hearing Healthcare Ecosystem

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A unified digital platform connecting every stakeholder in the hearing healthcare journey — patients, clinicians, suppliers, educators, and philanthropists — across the globe.

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1. System Overview

AudSpeX is a comprehensive, multilingual, cross-border web platform architected for the global hearing healthcare ecosystem. It bridges patients, hearing healthcare professionals, device manufacturers, suppliers, educational institutions, and charitable organizations under one unified digital roof.

1.1 Core Pillars

Pillar	Purpose
Connect	Patients discover audiologists, ENTs, speech therapists, and clinics near them
Commerce	Buyers and sellers transact hearing devices, tools, and accessories with intelligent dual-price logic
Care	Online diagnostic testing (PTA, Tympano, ASSR, BERA) through integrated audiology software
Community	Job board, education portal, scholarship opportunities, and donation ecosystem
Compassion	Zakat-eligible patient support, device bank, and donor-patient matching

1.2 Design Philosophy

- **Role-first architecture:** Every feature derives from the user's declared role(s)
 - **Dual-economy engine:** Separate pricing pipelines for retail consumers and trade buyers
 - **Privacy by design:** Patient data is encrypted, consent-gated, and blockchain-audited
 - **Global-first:** Multilingual, multi-currency, multi-country from day one
-

2. User Roles & Account Architecture

2.1 Role Taxonomy

Code	Role	Definition
HU	Hearing Device User	End-user wearing or needing a hearing aid
AU	Audiologist	Licensed audiology professional
AM	Audiometrist	Hearing test technician
ST	Speech Therapist	Speech-language pathologist
HT	Hearing Test Seeker	Person seeking a diagnostic hearing test
BY	Buyer	Purchases hearing devices and tools (includes audiologists)
SL	Seller	Lists and sells hearing devices, machines, and tools
RE	Repairer	Hearing device mechanic or repair technician
CO	Company	Hearing aid manufacturer or distributor
DO	Donor	Individual or organization wishing to donate
ES	ENT Surgeon	Ear, Nose, and Throat surgeon
OT	Occupational Therapist	Occupational therapy professional
BT	Behaviour Therapist	Behavioural therapy professional
SE	Special Educator	Special education professional
IN	Institution	Hospital, clinic, university, or institute
ZA	Zakat Patient	Financially disadvantaged patient seeking aid

A single user can hold multiple roles. For example, an Audiologist (AU) can also be a Buyer (BY) and simultaneously a Seller (SL).

2.2 Account Tiers

Tier	Roles Included	Privileges
Standard	HU, HT, DO, ZA, ST, OT, BT, SE, ES, RE	Browse, book tests, apply for jobs, donate
Buyer	AU (buying mode), BY	Purchase devices at buyer-tier pricing
Seller	SL, CO, AU (selling mode), RE	List products, access seller dashboard
Institutional	IN	Post jobs, list training programmes, offer scholarships

3. Registration & Profile System

3.1 Universal Registration Fields

All users must complete these fields at signup:

Field	Type	Constraints
User ID	Auto	HL-{YYMMDD}-{6-digit-seq}
Registered On	Auto	UTC timestamp
First Name	Text	2—50 characters
Last Name	Text	2—50 characters
Profile Picture	Image	5 MB max, JPG / PNG / WebP
Date of Birth	Date	Valid calendar date
Age	Auto-calculated	Derived from DOB
Country	Dropdown	ISO country list
State / Province	Dropdown	Filtered by country
City	Dropdown	Filtered by state
Address Line 1	Text	Required
Address Line 2	Text	Optional
Email	Email	Unique, OTP-verified
Phone	Phone	Unique, SMS-verified
Username	Text	6—30 chars, alphanumeric + underscore
Password	Password	Minimum 8 chars, 1 uppercase, 1 number, 1 special
Purpose / Role(s)	Multi-select	At least one role required

3.2 Role-Specific Extended Profiles

When a user selects a role, supplementary fields appear:

Hearing Device User (HU)

Field	Type
Current Device Name	Text
Current Device Model	Text
Using Since	Date
Previous Devices	Multi-entry list

Field	Type
Hearing Loss Degree	Dropdown (Mild / Moderate / Severe / Profound)
Affected Ear(s)	Checkbox (Left / Right / Both)

Audiologist / Audiometrist / Speech Therapist / Behaviour Therapist / Occupational Therapist / Special Educator (AU / AM / ST / BT / OT / SE)

Field	Type
Practice Type	Radio (Own Clinic / Employed)
Clinic / Institution Name	Text
Clinic / Institution Address	Textarea
Year of Qualification	Year
Degree / Qualification	Text
Awarding Institute	Text
Licence / Registration No.	Text
Specialties	Multi-select
Available Facilities	Multi-select
Services Offered	Multi-select
CV / Resume	File (PDF, 10 MB max)
Available Days & Timings	Day + Time slots

Hearing Test Seeker (HT)

Field	Type
Preferred Test Location	Text
Old Reports	File upload (multi, PDF / Image)
Test(s) Required	Multi-select (PTA / Tympano / ASSR / BERA / OAE / SISI / Free Field)
Currently Uses Hearing Aid?	Yes / No
If Yes → Device Name & Model	Text
If Yes → Using Since	Date

Buyer (BY) — includes Audiologists in buying capacity

Field	Type
Full / Business Name	Text

Field	Type
Business Registration No.	Text (optional)
Distributor of any hearing device company?	Yes / No
If Yes → Company Name(s)	Multi-entry
Total Companies Represented	Number
List All Companies	Textarea
Experience Letter / Certificate	File
<i>Status</i>	Pending Admin Approval

Seller (SL)

Field	Type
Full / Business Name	Text
Business Registration No.	Text
Business Address	Textarea
Years in Business	Number
Product Categories	Multi-select
Experience Letter	File
<i>Status</i>	Pending Admin Approval

Repairer (RE)

Field	Type
Skill Description	Textarea
Years of Experience	Number
Certifications	File
Device Types Repaired	Multi-select
Service Area	Text

Company (CO)

Field	Type
Company Name	Text
Registration / NTN No.	Text

Field	Type
Country of Registration	Dropdown
Has Local Distributor in Pakistan?	Yes / No
If Yes → Distributor Name & Contact	Text
Product Portfolio	Textarea
Company Profile / Experience Letter	File

ENT Surgeon (ES)

Field	Type
Hospital / Clinic Name	Text
Hospital / Clinic Address	Textarea
Degree (MBBS, FCPS, etc.)	Text
Year of Degree	Year
Total Experience (years)	Number
Specialisations	Multi-select
Licence No.	Text
Available Days & Timings	Day + Time slots

Donor (DO)

No supplementary fields required. Optional preferences:

- Donation type (Equipment / Money / Both)
- Anonymous donor toggle

Institution (IN) — Hospital / University / Institute / NGO

Field	Type
Institution Name	Text
Institution Type	Dropdown
Registration No.	Text
Address	Textarea
Website	URL
Contact Person Name	Text
Contact Person Designation	Text

4. Three-Tier Account Model & Marketplace Engine

4.1 Account Tier Matrix

AudSpeX PLATFORM	
STANDARD (Normal Users) • Browse listings • View Price B • Request tests • Apply for jobs • Donate	BUYER (Approved) • Browse listings • View Price A • Purchase at lower price • Bulk orders
SELLER (Approved) • List products with base price • Manage inventory • View sales analytics • Can also buy (buyer privileges)	

4.2 Seller Product Listing Flow

1. Seller logs in and navigates to **Seller Dashboard** → **Add Product**

2. Enters: product name, category (Hearing Aid / Battery / Tool / Accessory / Machine / Part), brand, model, condition (New / Used / Refurbished), description, **base price**, quantity, images

3. System **automatically applies +10%** to base price — this becomes the Buyer Price

4. Submission enters **Admin Approval Queue** — not visible to public until approved
- 11 / 42

5. Dual-Price Approval Algorithm

5.1 The Problem

- **Standard users** (end consumers) should see a reasonable retail price
- **Buyers** (audiologists, clinics, bulk purchasers) should see a wholesale price
- **Sellers** want one base price but need to reach both markets without undercutting themselves

5.2 Algorithm Flow

INPUT: Seller submits product with BASE_PRICE

PROCESS:

Step 1: System calculates Buyer Price = BASE_PRICE × 1.10

Step 2: Two entries created in Admin Approval Queue:

ENTRY A – Standard User Tier:

Listed Price = Buyer Price × (1 + MARGIN_STANDARD%)

Margin = configurable by admin (default 15–30%)

ENTRY B – Approved Buyer Tier:

Listed Price = Buyer Price (only the 10% uplift)

No additional margin

Step 3: Admin reviews both entries with full editing capability

OUTPUT: Two distinct prices for two distinct audiences

5.3 Admin Approval Dashboard

Feature	Description
Dual-Price Preview	Both prices displayed side by side
Price Override	Admin can manually set any price for either tier
Margin Calculator	Real-time profit margin display for both tiers
Price History Log	Complete audit trail with timestamps and admin IDs
Bulk Approval	Select and approve multiple products simultaneously
Change Reason	Admin must log a reason for any price override
Scheduled Publishing	Set a future date for automatic listing activation

5.4 Approval Screen Layout

PRODUCT: Phonak Audéo Lumity L90
Seller: ABC Hearing Aids – Karachi

Base Price (by Seller): Rs. 85,000
Auto +10% (Buyer Price): Rs. 93,500

STANDARD TIER	BUYER TIER
Rs. 118,000	Rs. 93,500
Margin: 26.8%	Margin: 10%
[Adjust -] [+] +	[Adjust -] [+]
[✓ Approve Both] [Modify] [Reject]	
Reason for change: _____	

6. Zakat / Financial Aid Patient Registration

A dedicated registration portal for financially disadvantaged patients who are Zakat-eligible (Mustahiq), low-income government employees (Safaid Posh), unemployed, or below the poverty line.

6.1 Registration Fields

Field	Type
Full Name	Text
National ID (CNIC)	Text, encrypted
Monthly Income	Number
Income Certificate	File
Number of Dependents	Number
Current Hearing Device (if any)	Text
Device Model	Text
Using Since	Date
Latest Audiological Report	File
Type of Need	Multi-select
• New Hearing Device	Checkbox
• Device Parts / Repair	Checkbox
• Hearing Test	Checkbox
• Speech Therapy	Checkbox
• Battery / Accessories	Checkbox
Detailed Description of Need	Textarea
Doctor's Prescription / Referral	File, required

6.2 Privacy & Verification Protocol

- All data is encrypted at rest (AES-256) and in transit (TLS 1.3)
 - Every application is manually verified by a Zakat/Donation Admin before publication
 - Applicants receive a unique tracking ID to monitor their application status
 - No personal contact information is visible to donors — only anonymised needs
-

7. Donor Platform (National & International)

7.1 Donor Classification

Type	Description
Individual Donor	Personal donors, national or international
Institutional Donor	NGOs, charitable trusts, foundations
Corporate Donor	Companies with CSR programmes

7.2 Donation Workflow

1. Verified Zakat patient needs are displayed anonymously (no name, photo, or contact)
2. Each need shows: age range, gender, required item/service, estimated cost, request age
3. Donor clicks **"I Want to Donate This"**
4. Donor is prompted to contact the AudSpeX institute (platform admin) via:
 - Platform messaging system
 - Email notification to admin
5. Admin mediates the donation — matching the right donor to the right patient
6. Donor receives:
 - Tax receipt / acknowledgment
 - Impact report (with patient consent)
 - Verified donor badge on profile

7.3 International Donation Support

- Multi-currency display: USD, GBP, EUR, AED, SAR, PKR
 - Live exchange rate indicators
 - International wire transfer instructions
 - Planned: PayPal / Stripe integration for online giving
-

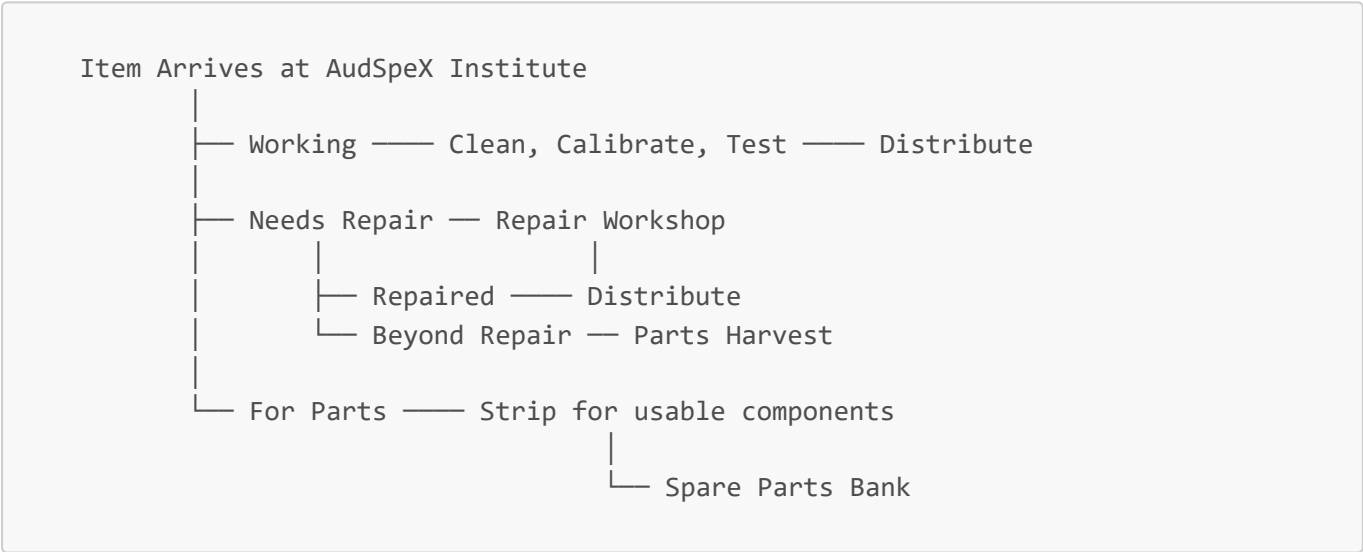
8. Hearing Device & Equipment Bank

A recycling-and-distribution programme that gives unused audiological equipment a second life.

8.1 Accepted Donation Categories

Category	Examples
Hearing Aids	BTE, ITE, CIC, RIC — any brand or model
Audiological Machines	Audiometers, tympanometers, OAE, BERA, ASSR systems
Hearing Aid Tools	Cerumen removal kits, probes, analysers
Parts & Accessories	Domes, tubes, ear moulds, batteries, chargers
Testing Equipment	Sound booths, headphones, couplers
Any audiology-related item	Working or non-working

8.2 Donation Processing Pipeline



8.3 Equipment Bank Dashboard (Admin View)

HEARING DEVICE BANK				
Total Items Donated:		1,247		
Items Distributed:		1,098		
Under Repair:		89		
Awaiting Collection:		60		
ID	Item	Condition	Status	Recipient
01	RIC	Working	Given	Anon#042
02	BTE	Needs Rpr	Shop	—

03	ITE	For Parts	Bank	-
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9. Job Portal

9.1 Employer Posting Form

Field	Details
Job Title	Text
Department	Dropdown (Audiology / Speech Therapy / ENT / Admin / Other)
Employment Type	Full-time / Part-time / Contract / Internship / Remote
Location	Country → State → City
Salary Range	Min — Max
Currency	Dropdown
Job Description	Rich text
Requirements	Rich text
Benefits	Rich text
Application Deadline	Date
Contact Email	Email
Contact Phone	Optional
Scope	National / International

9.2 Job Seeker Features

- Browse by category, location, salary range
- One-click apply — CV attached automatically from profile
- Application status tracking
- Platform-mediated direct messaging with employers (identity masked until both parties agree)
- Saved job alerts
- Recommended jobs based on user role and profile keywords

9.3 Intelligent Job Matching

```
Score = (CategoryMatch × 0.4) + (LocationMatch × 0.3)  
       + (LevelMatch × 0.2) + (KeywordMatch × 0.1)
```

Jobs sorted by descending score

10. Education & Training Portal

10.1 Programme Posting Form

Field	Details
Programme Title	Text
Provider Name	Auto-filled from institution profile
Programme Type	Degree / Diploma / Certificate / Workshop / Seminar / Short Course
Mode	In-Person / Online / Hybrid
Start Date	Date
End Date	Date
Duration	Text
Curriculum / Syllabus	File + Rich text
Fee	Number
Currency	Dropdown
Eligibility Criteria	Rich text
Application Process	Rich text
Scope	National / International

10.2 Learner Features

- Browse all programmes with filters by type, mode, location, fee
 - Direct enrolment through the platform
 - Bookmark and compare programmes
 - Training calendar view
 - Certificate upload after completion (added to profile automatically)
-

11. Scholarship Portal

11.1 Scholarship Posting Form

Field	Details
Scholarship Name	Text
Provider Name	Text
Provider Type	Institution / NGO / Government / Corporate
Field of Study	Multi-select
Amount / Coverage	Text
Number Available	Number
Eligibility Criteria	Rich text
Application Requirements	Rich text
Application Deadline	Date
Apply Link / Contact	Text
Scope	National / International

11.2 Applicant Features

- Browse available scholarships
- Filter by field, amount, deadline, region
- Apply directly via provided link or platform-integrated form
- Track application status from the dashboard

12. Location-Based Provider Directory

12.1 Directory Content by Provider Type

Provider Type	Information Displayed
Audiologist	Name, clinic, address, timings, services, contact
ENT Surgeon	Hospital, OPD timings, days available, qualifications
Speech Therapist	Clinic, availability, specialities
Audiometrist	Testing facility, equipment available
Hearing Aid Seller	Shop location, brands stocked
Repairer	Service area, skills, turnaround time
Clinic / Hospital	All departments, doctors, timings

12.2 Search and Filter Capabilities

- **Quick search** — "Audiologist near me"
- **Advanced filters:**
 - Provider type
 - Country → State → City
 - Day of availability
 - Time slot (Morning / Afternoon / Evening)
 - Service type (PTA / Hearing Aid Fitting / Paediatrics / Cochlear Implants)
 - Gender preference
 - Language spoken
- **Map view** — Interactive map with pinned provider locations
- **List view** — Card layout with key information and rating
- **Radius search** — 5 km / 10 km / 25 km / 50 km

12.3 Provider Self-Management Dashboard

Each registered provider can:

- Add and update their listing
 - Mark available and unavailable days
 - Set consultation hours
 - Update services
 - Respond to patient enquiries
 - View profile visit statistics
-

13. Online Diagnostic Test Module

13.1 Supported Tests

Test	Category	Description
Pure Tone Audiometry (PTA)	Behavioural	Air conduction, bone conduction, MCL, UCL
Tympanometry	Objective	Middle ear function, compliance, pressure, gradient
Acoustic Reflex (ASSR)	Objective	Brainstem response to sound
BERA / ABR	Objective	Auditory brainstem response
OAE (DPOAE / TEOAE)	Objective	Cochlear outer hair cell function
SISI	Behavioural	Speech discrimination in noise
Free Field Testing	Behavioural	Hearing assessment without headphones

13.2 End-to-End Workflow

1. Patient books a test via the platform (selects test type, preferred audiologist, date, and time)
2. Audiologist receives an instant booking notification
3. At the scheduled time, the audiologist uses the integrated test software:
 - Connects to audiometer via USB or Bluetooth (clinic setup)
 - Or uses calibrated software-based testing with validated headphones (remote setup)
4. Test is conducted; results stream in real time
5. Audiogram or tympanogram graph is auto-generated
6. Results are saved to the patient's profile on AudSpeX
7. Patient receives:
 - Digital report (downloadable PDF)
 - Hearing classification (Normal / Mild / Moderate / Severe / Profound)
 - Hearing type (Conductive / Sensorineural / Mixed)
 - Clinical recommendations
8. Tele-rehabilitation follow-up via platform video call

13.3 Auto-Generated Report Layout

AUDIOLOGICAL EVALUATION REPORT	
Patient ID:	HL-240615-003842
Name:	[Patient Name]
Age / Gender:	45 / Male
Test Date:	15-June-2026
Performed By:	Dr. [Audiologist Name]
Facility:	[Clinic / Hospital Name]
RIGHT EAR LEFT EAR	
PTA + Classification	PTA + Classif.

Type: Sensorineural	Type: Mixed
AC Mean: 62.5 dB	AC Mean: 71.3 dB
BC Mean: 28.8 dB	BC Mean: 35.0 dB
Tympano: Type Ad	Tympano: Type As
Compliance: 0.6 mmho	Compliance: 0.3

Impression: Moderate to Severe Sensorineural Hearing Loss (Right); Mild to Moderate Mixed Hearing Loss (Left).

Recommendation: Bilateral hearing aid fitting.
Refer for medical clearance.

Generated by AudSpeX Audiology Suite

14. Algorithms & Intelligence Engine

Each algorithm is presented with a defined input, step-by-step processing logic, and a clearly stated output. These specifications are ready for direct translation into production code.

Algorithm 14.1 — AI-Powered Hearing Symptom Checker

INPUT:

- Patient responses from an interactive chatbot questionnaire
- Data points: age group, symptoms, duration, noise exposure history, medical history

PROCESS:

1. Collect responses from chatbot conversation.
2. Map each symptom to a weighted decision node:
 - "Difficulty hearing in noise" → Sensorineural score +3
 - "Ear fullness or pain" → Conductive score +4
 - "Sudden onset within 72 hours" → FLAG = EMERGENCY → route to SOS
 - "Gradual onset over 6+ months" → Age-related score +2
3. Calculate aggregate risk score per condition category.
4. If any risk score exceeds the defined threshold:
 - Recommend the appropriate specialist (Audiologist or ENT)
 - Suggest the relevant test type (PTA / Tympano / BERA)
 - Provide an estimated severity range.
- Else:
 - Recommend monitoring with hearing hygiene advice.
5. Offer a one-click appointment booking with the matched specialist.
6. Log anonymized data for population health analytics.

OUTPUT:

Risk assessment + Specialist recommendation + Appointment booking link

Algorithm 14.2 — Device Recommendation Engine

INPUT:

- Patient audiogram – AC thresholds at 250, 500, 1k, 2k, 4k, 8k Hz
- Lifestyle profile – at-home / active / professional / musician
- Budget range – Low / Mid / High / Premium
- Previous device experience – brand, model, satisfaction score

PROCESS:

1. Classify hearing loss severity:
$$PTA = (AC_{500} + AC_{1k} + AC_{2k} + AC_{4k}) / 4$$
 - $PTA \leq 25$ dB → Mild → Category A
 - PTA 26–40 → Moderate → Category B
 - PTA 41–60 → Severe → Category C
 - $PTA > 60$ → Profound → Category D

2. Match category to device power range:
 - Category A → Basic BTE, RIC
 - Category B → Mid-power BTE, ITE, RIC
 - Category C → High-power BTE, RIC
 - Category D → Super-power BTE, CROS, or CI candidacy check
3. Filter by lifestyle feature requirements.
4. Filter by budget tier.
5. Rank remaining devices by a compatibility score (0–100) derived from feature match percentage.
6. Return the top 5 devices with brand, model, estimated price, and compatibility score.

OUTPUT:

Ordered list of recommended devices + Compatibility score + Reasoning

Algorithm 14.3 — Universal Patient Passport

INPUT:

- Patient demographic data and complete audiological history
- All test results, device records, prescriptions, clinical reports
- Patient consent preferences – who may view which data types

GENERATION PHASE:

1. Collect all patient data from the unified database.
2. Hash each record into a SHA-256 block.
3. Link blocks into a patient-specific Merkle tree.
4. Deploy a smart contract on-chain containing:
 - Patient ID (hashed)
 - Timestamp for each record
 - RBAC access control rules
5. Generate a QR code encoding the blockchain pointer and access token.
6. Deliver to the patient:
 - Printable QR code + mobile wallet
 - Master password / biometric authentication
 - Emergency access PIN for hospital use

ACCESS PHASE:

1. Provider scans the QR code.
2. Provider authenticates via platform role-based access.
3. Patient receives a real-time notification:

"Dr. [Name] is requesting access to your records."
4. Patient approves or denies the request.
5. If approved → a temporal access token is granted (valid for 24 hours).
6. Provider views complete hearing history in read-only mode.
7. Every access event is logged to the blockchain for audit.

REVOCATION:

- Patient revokes access → Token invalidated immediately.
- 24-hour expiry → Token auto-expires.

OUTPUT:

Immutable, patient-controlled, portable hearing health record

Algorithm 14.4 — Remote Hearing Aid Tuning (Tele-Fitting)

INPUT:

- Patient hearing aid serial number, brand, and model
- Current audiogram (verified within the last 6 months)
- Patient subjective feedback – 7 categorical questions
- Target gain prescription – NAL-NL2 or DSL v5

PROCESS:

1. Verify patient identity via two-factor authentication.
2. Establish a secure WebRTC session between audiologist and patient.
3. Audiologist initiates a remote programming session:
 - a. Authenticate with the hearing aid manufacturer's API.
 - b. Establish Bluetooth LE connection to the device.
 - c. Read current firmware version.
 - d. Pull current parameter settings – gain, output, compression, channels.
4. Compare current settings against the target prescription:
 - Calculate deviation per frequency band (250 Hz through 6 kHz).
 - If deviation exceeds 5 dB in any band → flag for adjustment.
5. Audiologist adjusts parameters in real time:
 - Real Ear Measurement (REM) if a coupler is available.
 - Or simulated REM based on age, earmold, and venting.
 - Patient provides live feedback: "Too loud / Too soft / Better."
6. Run verification checks:
 - Compute predicted Speech Intelligibility Index (SII).
 - Validate against target SII (minimum 0.6 for moderate loss).
 - Run acoustic feedback path simulation.
7. If all checks pass:
 - Finalise and write settings to the device.
 - Generate a before/after adjustment report.
- Else:
 - Flag the case for a physical follow-up.
8. Log the session – timestamp, audiologist ID, parameters changed, patient feedback – encrypted end-to-end.

OUTPUT:

Successfully tuned hearing aid + Session report + Follow-up schedule

Algorithm 14.5 — Emergency Hearing Alert (SOS) Dispatch

INPUT:

- Patient SOS trigger – app button press, voice command, or fall detection
- GPS coordinates of the patient
- Pre-stored medical profile – allergies, medications, history, blood type

PROCESS:

1. Classify priority based on trigger type:
 - Manual SOS button → HIGH
 - Sudden hearing loss form → HIGH
 - Fall detection → CRITICAL
 - Tinnitus spike → MEDIUM
2. Locate the nearest three providers using a geospatial query:


```
SELECT * FROM providers
WHERE type IN ('ENT', 'Audiologist') AND is_available = TRUE
AND GeoDistance(location, :patient_location) < 50 km
ORDER BY GeoDistance ASC LIMIT 3
```
3. Dispatch simultaneously to the top 3 matches:
 - Push notification (high-priority)
 - SMS with patient name, age, and GPS link
 - Email with a detailed medical summary
4. Send confirmation to the patient:
 - "Help is on the way. Dr. [Name] has been notified."
 - Estimated response time.
 - Nearest hospital with an audiology department.
5. If no provider responds within 5 minutes:
 - Escalate to the nearest hospital emergency department.
 - Route the patient via maps integration.
 - Notify the pre-registered emergency contact.
6. If a provider accepts:
 - Open a secure video or audio channel.
 - Provider accesses limited emergency medical data.
 - Track response time for quality metrics.
7. Close the loop:
 - Patient rates the response quality.
 - Provider logs the outcome – diagnosis, referral, or discharge.
 - Data feeds into the platform quality score.

OUTPUT:

Rapid-response alert + Provider match + Escalation protocol

Algorithm 14.6 — Hearing Device Lifecycle Prediction

INPUT:

- Device serial number → lookup: brand, model, manufacture date
- Usage patterns – hours per day from patient app telemetry
- Environmental factors – humidity and temperature via weather API
- Previous service records
- Patient-reported issues from the app

PROCESS:

1. Determine baseline expected lifespan from manufacturer data:
 - BTE → 5 years ± 1.2
 - RIC → 4 years ± 1.0
 - ITE → 4 years ± 1.5

- CIC → 3 years ± 1.0
- 2. Adjust for usage intensity:
Usage factor = actual hours per day ÷ 16
 - > 1.2 → predicted life × 0.85
 - < 0.6 → predicted life × 1.10
- 3. Adjust for environmental stress:
 - Humidity > 70 % + temp > 30 °C → factor = 0.80
 - Humidity < 40 % + temp < 20 °C → factor = 1.05
 - Default → factor = 1.00
- 4. Compute predicted remaining life:
Remaining days = (manufacture date + adjusted lifespan) - today
- 5. Generate alerts:
 - < 90 days → "Replace soon" alert + trade-in offer
 - < 30 days → "Critical – schedule replacement"
 - Service overdue → Maintenance alert
 - Battery cycles > 300 → Battery replacement suggestion

OUTPUT:

Remaining life estimate + Tiered alerts + Trade-in valuation

Algorithm 14.7 — Tinnitus Severity Scoring

INPUT:

- Patient self-reported data from a Tinnitus Tracking Diary:
 - Frequency (daily occurrence) — scale 0-10
 - Loudness level — scale 0-10
 - Sleep disturbance — scale 0-10
 - Anxiety or distress level — scale 0-10
 - Duration of condition — days / months / years
 - Identified trigger factors — multi-select

PROCESS:

1. Calculate the Tinnitus Severity Index (TSI):

$$\text{TSI} = (\text{Frequency} \times 0.25) + (\text{Loudness} \times 0.20) + (\text{Sleep} \times 0.25) + (\text{Anxiety} \times 0.20) + (\text{Duration factor} \times 0.10)$$

Duration factor:

 - < 3 months → 2
 - 3-12 months → 4
 - 1-5 years → 6
 - > 5 years → 8
2. Classify severity:
 - TSI ≥ 8.0 → SEVERE → Urgent ENT + Audiology referral
 - TSI ≥ 5.0 → MODERATE → Sound therapy + counselling
 - TSI ≥ 3.0 → MILD → Masking sounds + sleep hygiene
 - TSI < 3.0 → MINIMAL → Reassure and monitor
3. Match interventions to trigger profile:

- Stress → Add Cognitive Behavioural Therapy (CBT)
- Noise → Hearing protection counselling
- Medication → Flag for medical review
- Jaw pain → Suggest TMJ evaluation

4. Generate a personalised sound therapy playlist from the library.
5. Schedule a follow-up in 14 days, repeat TSI, and adjust.

OUTPUT:

Tinnitus Severity Index + Severity level + Personalised intervention plan

Algorithm 14.8 — Smart Inventory Forecasting

INPUT:

- Historical sales data – last 12 months
- Current stock levels per SKU
- Upcoming seasons or events – e.g., World Hearing Day
- Supplier lead time in days
- Marketing campaign schedule

PROCESS:

1. Decompose the time series:
 - Trend component = 30-day moving average
 - Seasonal factor = same month last year ÷ average
 - Residual = actual – (trend × seasonal)
2. Compute forecast for each SKU:
Forecast = Trend_next × Seasonal_current × Growth_factor
3. Calculate reorder point:
Safety_stock = (max daily sales × max lead time)
– (avg daily sales × avg lead time)
Reorder_point = (avg daily sales × avg lead time) + Safety_stock
4. Generate stock alerts:
 - Stock ≤ Reorder_point → "Order within 7 days"
 - Stock = 0 → "Out of stock – urgent reorder"
 - Stock > Reorder_point × 3 → "Overstock – consider promotion"
5. Suggest order quantity:
Order = Forecast – Current_stock + Safety_stock
6. If auto-order preference is enabled → generate and send purchase order.
Otherwise → send notification with suggested order.

OUTPUT:

Reorder alerts + Optimal order quantities + Sales trend dashboard

Algorithm 14.9 — Audiologist-Patient Matching

INPUT:

- Patient preferences:

- Location and search radius
- Language, gender preference
- Available days and times
- Required specialisation
- Insurance or payment method
- Minimum rating threshold
- Provider profiles with matching attributes

PROCESS:

1. Apply hard-constraint filters:
 - Location radius query
 - Specialisation match
 - Day and time overlap
 - Accepted payment method
2. Compute a weighted compatibility score:
 - Score = 0
 - +25 if language matches
 - +10 if gender matches
 - +10 × provider rating
 - +1.5 × max(0, 30 - distance in km)
 - +0.5 × years of experience
 - +15 if fee is within budget
3. Rank providers by descending score.
4. Return the top 5 with profile summaries, score breakdown, and available slots.
5. Patient selects → booking is initiated → provider is notified.

OUTPUT:

Ranked list of best-matched providers + Compatibility explanation

Algorithm 14.10 — Speech-in-Noise (SIN) Test Automation

INPUT:

- Patient audiogram – AC thresholds
- Patient age
- Native language
- Test environment – quiet clinic or remote with calibrated headphones

PROCESS:

1. Determine test configuration:
 - Clinic → QuickSIN, WIN, or BKB-SIN tracks
 - Remote → Platform-validated digit-in-noise test
2. Present 10 sentences per SNR condition:
 - SNR levels: +25, +20, +15, +10, +5, 0, -5 dB
 - Each sentence contains 5 key words.
 - Patient repeats; system records correct and incorrect words.
3. Score each SNR level as a percentage.
4. Compute SNR-50 – the signal-to-noise ratio at which 50 % of words are correctly identified – by interpolation.
5. Classify:

- SNR-50 < 0 dB → Excellent (normal)
 - SNR-50 0–3 dB → Good (slight difficulty)
 - SNR-50 3–6 dB → Moderate difficulty
 - SNR-50 6–9 dB → Significant difficulty
 - SNR-50 > 9 dB → Severe difficulty
6. If SNR-50 exceeds 3 dB, recommend hearing aids with:
- Digital noise reduction
 - Directional microphones
 - Speech enhancement algorithms

OUTPUT:

SNR-50 score + Classification + Device feature recommendations

Algorithm 14.11 — Hearing Deterioration Trend Analysis

INPUT:

- Minimum 2 historical audiograms per patient (4+ per year recommended)
- AC + BC thresholds at standard frequencies
- Test dates

PROCESS:

1. Align audiograms by frequency; interpolate any missing points.
2. For each frequency, compute the annual rate of change:

$$\text{Rate} = (\text{current threshold} - \text{previous threshold}) \div (\text{days between tests} \div 365.25)$$
3. Flag significant deterioration:
 - > 5 dB/year at any single frequency → Rapid deterioration
 - > 3 dB/year at 3 or more frequencies → Consistent pattern
 - Negative rate (improvement) → Possible measurement error
4. Predict the threshold at the next visit (3 months out):

$$\text{Predicted} = \text{current threshold} + (\text{rate} \times 0.25)$$
5. Generate a clinical alert:
 - Rapid → "Immediate medical referral: rule out acoustic neuroma, Meniere's disease, autoimmune inner ear disease, ototoxicity"
 - Consistent → "Progressive loss detected. Adjust hearing aid settings. Schedule follow-up."
6. Update the patient dashboard with a trend graph – actual thresholds overlaid with the predicted trajectory, colour-coded green (stable), yellow (monitor), red (urgent).

OUTPUT:

Deterioration rate per frequency + Early warning + Predicted trajectory

Algorithm 14.12 — Cochlear Implant Candidacy Predictor

INPUT:

- Audiogram in the severe-to-profound range

- Speech recognition score – AzBio or CNC words in quiet
- Duration of hearing loss in years
- Patient age
- Previous hearing aid trial – model, duration, benefit level
- Medical contraindications – CT or MRI findings

PROCESS:

1. Audiology criteria:
 - PTA > 70 dB both ears → +30
 - AzBio < 50 % in best-aided → +30
 - AzBio 50-60 % → +15
 - AzBio > 60 % → +0 (not a candidate)
2. Duration criteria:
 - < 5 years → +15
 - 5-10 years → +10
 - > 10 years → +5
3. Age criteria:
 - < 5 years → +20
 - 5-18 years → +15
 - 18-65 years → +10
 - > 65 years → +5
4. Device trial history:
 - Failed trial (no benefit) → +15
 - Partial benefit → +5
 - No trial → Flag: "Recommend 3-month HA trial first"
5. Medical clearance:
 - Contraindications present → Reject (refer for medical management)
 - Clear → Proceed
6. Calculate total:

Total = Audiology + Duration + Age + Device_trial
7. Classify:
 - ≥ 70 → Strong candidate – proceed with CI evaluation
 - 50-69 → Borderline – consider additional testing
 - < 50 → Not currently a candidate – continue hearing aid use

OUTPUT:

Candidacy score + Classification + Clinical recommendation

Algorithm 14.13 — Newborn Hearing Screening Tracker

INPUT:

- Date of birth
- OAE or BERA screening results from the hospital
- Risk factors – NICU stay > 5 days, family history, infections
- Follow-up appointment history

PROCESS:

1. Record the initial screening:
 - OAE pass in both ears → Schedule 6-month re-screen
 - OAE refer in one or both ears → Schedule 1-month follow-up BERA

2. Track follow-up compliance:
 - Completed → Record results
 - Missed → Send SMS + email + app reminder
 - 2nd missed → Flag for social worker or community health visit
3. On follow-up BERA:
 - < 25 dB → Normal hearing → Discharge
 - 26-40 dB → Mild loss → Hearing aid fitting + monitor
 - 41-60 dB → Moderate loss → Hearing aid + speech therapy
 - 61-80 dB → Severe loss → Hearing aid + CI evaluation
 - > 80 dB + age < 12 months → CI referral
4. Generate developmental milestone tracker at 6, 12, 18, 24, 36 months.
5. Flag if milestones are not met.
6. Contribute anonymised data to the national screening registry.

OUTPUT:

Screening record + Follow-up schedule + Milestone tracker + Registry data

Algorithm 14.14 — IoT Hearing Aid Performance Monitor

INPUT:

- IoT-enabled hearing aid with telemetry capability
- Streaming data points every 15 minutes during use:
 - Battery voltage
 - Volume control position
 - Feedback cancellation activity level
 - Usage hours and environment classification
 - Acoustic environment classifier output (quiet / noise / music / wind)

PROCESS:

1. Connect to the device via manufacturer API or MQTT bridge.
2. Establish a secure data stream with TLS 1.3.
3. For each data batch, run four health checks:

Battery check:

- < 1.1 V → "Critical – replace immediately"
- 1.1-1.2 V → "Low – replace within 3 days"
- > 300 cycles → "Aging – plan replacement"

Feedback check:

- Activity > 50 % → "Possible feedback loop – check earmold fit, verify gain, clean microphone ports"

Usage check:

- < 4 hours/day → "Low compliance – patient may need counselling"
- > 16 hours/day → "High dependency – verify battery adequacy"

Environment check:

- Mostly noise → "Verify noise reduction is active"
- Mostly quiet → "Consider telecoil for TV or phone"

4. If any check fails → send an automated alert:
 - Patient: simple-language notification with suggested action
 - Audiologist: technical report with recommended adjustment
5. Aggregate a weekly performance report:
 - Uptime percentage, average daily use, battery trend, feedback events, environment distribution, patient satisfaction score.

OUTPUT:

Real-time device telemetry + Automated health alerts + Weekly report

Algorithm 14.15 — Fraud Detection for Seller Reviews

INPUT:

- Review text content
- Reviewer profile – account age, verified purchase, role
- Star rating (1–5)
- Timestamp
- IP address and device fingerprint
- Historical review patterns of the reviewer

PROCESS:

1. Verify purchase:
 - No purchase → Reject review.
 - Purchased but returned → Flag for manual review.
2. Check review velocity:
 - > 3 reviews in 24 hours → Flag
 - > 10 reviews in 7 days → Flag
3. NLP sentiment analysis:
 - Detect patterns typical of fake reviews:
 - superlatives without specifics, identical text on multiple products.
 - Score -20 if suspicious.
4. Check timing coordination:
 - Multiple 5-star reviews within the same hour → Score -30
 - Multiple 1-star reviews from the same IP range → Score -30
5. Compute trust score (starting at 100):
 - Subtract penalties from steps 2, 3, and 4.
6. Decision:
 - < 50 → Reject review and flag account for investigation
 - 50–69 → Publish with "Unverified" badge
 - ≥ 70 → Publish normally
7. If the same reviewer is flagged 3 or more times → suspend account.

OUTPUT:

Verified publication or Flagged for manual review

Algorithm 14.16 — Insurance / Billing Code Generator

INPUT:

- Service performed – PTA, Tympano, hearing aid fitting, etc.
- Time spent in minutes
- Complexity level – straightforward / moderate / complex
- Patient insurance provider
- Diagnosis code (ICD-10)
- Provider credentials

PROCESS:

1. Map the service to the appropriate CPT code:
 - PTA (air + bone) → 92557
 - PTA (air only) → 92552
 - Tympanometry → 92567
 - Acoustic reflex → 92568
 - BERA → 92585
 - ASSR → 92586
 - HA evaluation and fitting → 92590 + 92604
 - Real Ear Measurement → 92626
2. Determine modifiers based on complexity:
 - Time > 60 min + complex → -22 (increased service)
 - Bilateral → -50
 - Paediatric < 3 years → -59 (distinct service)
3. Validate against insurance bundling rules (NCCI edits).
4. Flag any code that requires prior authorisation.
5. Generate the billing output:
CPT code + modifiers, ICD-10 linkage, service date, provider NPI, and the pre-calculated charge amount.
6. If integrated with a practice management system → auto-submit.
Otherwise → generate a CMS-1500 form.

OUTPUT:

Complete billing code set + Ready-to-submit claim form

Algorithm 14.17 — Multi-Lingual Test Instruction Generator

INPUT:

- Patient's preferred language – selected during registration
- Test type – PTA / Tympano / ASSR / BERA / OAE
- Patient age group – child / adult / geriatric
- Hearing loss severity – to calibrate voice level

PROCESS:

1. Select the language from a library of 20+ supported languages.
2. Load test-specific instruction templates:
 - PTA: "You will hear beeps. Press the button every time you hear a sound, even if very faint."
 - Tympano: "You will feel pressure in your ear. Stay still. Do not swallow or speak."
3. Adapt for age:

- < 12 → Child-friendly: "Let's play a listening game! Press the button when you hear the bird."
 - > 65 → Large font + spoken instructions.
4. Generate audio instructions via text-to-speech in the patient's native language, with a choice of male or female voice.
 5. Display on-screen motivational prompts during the test.
 6. If no response for 30 seconds → repeat instructions in simpler language and offer live help via video call.

OUTPUT:

Native-language, age-adapted, test-specific instructions

Algorithm 14.18 — Gamified Auditory Learning Progression

INPUT:

- Child's age – 3 to 12 years
- Hearing loss configuration – audiogram
- Device type – HA / CI / Bimodal
- Baseline speech perception score
- Daily practice commitment in minutes

PROCESS:

1. Generate a personalised learning path with 5 levels:

Level 1 – Sound Detection (30 activities)	→ 90 % accuracy to pass
Level 2 – Sound Discrimination (40 activities)	→ 80 % accuracy
Level 3 – Closed-set Word Recognition (50)	→ 80 % accuracy
Level 4 – Open-set Word Recognition (60)	→ 70 % accuracy
Level 5 – Sentence Comprehension (70)	→ 70 % accuracy
2. Dynamically adjust difficulty:
 - Score > 80 % → Auto-advance to the next level.
 - Score < 50 % → Simplify by reducing noise or adding visual cues.
 - Score 50-80 % → Repeat with variation.
3. Reward system:
 - Correct answers earn "Hearing Stars."
 - 10 stars unlock a new game, skin, or character.
 - Daily streak gives bonus stars.
 - Level completion earns a printable digital certificate.
4. Parent dashboard:
 - Practice time (daily, weekly, monthly).
 - Accuracy trend chart.
 - Identified areas of difficulty.
 - Recommended home activities.
5. Weekly report for the audiologist:
 - Progress data to inform therapy adjustments.
 - Flag if no progress is made in 14 days.

OUTPUT:

Adaptive learning path + Performance analytics + Parent and clinical reports

Algorithm 14.19 — Tele-Audiology Session Management

INPUT:

- Appointment booking – patient, audiologist, test type, time
- Platform resources – bandwidth estimation, device compatibility
- Session type – consultation / test / follow-up / fitting

PRE-SESSION:

1. 24 hours before → Send reminders to both parties.
2. 30 minutes before → Check device readiness:
 - Patient: camera, microphone, internet speed.
 - Audiologist: software version, calibrations, peripherals.
3. 5 minutes before → Virtual waiting room opens.

DURING SESSION:

4. Establish WebRTC connection with adaptive bitrate:
 - < 1 Mbps → Audio-only + screen sharing.
 - 1–3 Mbps → 480p video + screen sharing.
 - > 3 Mbps → 720p/1080p video + screen sharing.
5. Record the session with consent:
 - Encryption: AES-256 at rest, TLS 1.3 in transit.
6. Activate live captioning for hearing-impaired patients:
 - Speech-to-text overlay translated into the patient's language.

POST-SESSION:

7. Generate a session summary:
 - Duration, tests performed, key discussion points, recommendations, follow-up schedule.
8. Collect patient feedback on session quality.
9. If quality rating < 3 → Flag for quality assurance review.
10. Store the encrypted recording for 7 years (medico-legal requirement).

OUTPUT:

Complete session log + Encrypted recording + Summary + QA score

Algorithm 14.20 — Hearing Health Research & Population Analytics

INPUT:

- Anonymised, opt-in aggregate data from the platform:
 - Demographics – age, region, gender (no PII)
 - De-identified audiometric test results
 - Device usage patterns
 - Treatment outcomes
 - Provider density data

PROCESS:

1. Apply differential privacy:
 - Add controlled noise to each data point.
 - Enforce k-anonymity – each group contains 5+ individuals.

2. Generate standard reports:

REPORT A – Prevalence Heatmap:

Hearing loss percentage by region, filterable by age group, severity, and type.

REPORT B – Market Intelligence:

Brand market share by region. Price trends over time (buyer vs. standard tier). Most popular categories.

REPORT C – Clinical Outcomes:

Average improvement in speech scores post-fitting.
Comparison across brands and models.
Impact of early intervention in paediatric cases.

REPORT D – Access and Equity:

Provider-to-population ratio by region.
Average distance to the nearest audiologist.
Disparity analysis by income level.

3. If a research institution requests custom data:

- Submit a research proposal for admin review.
- If approved → export in HIPAA Limited Data Set format.
- Data usage agreement must be signed before release.

4. Grant matching:

- Scan available research grants (NIH, WHO, local funding bodies).
- Match against platform data capabilities.
- Alert eligible institutions: "Your data can support this grant."

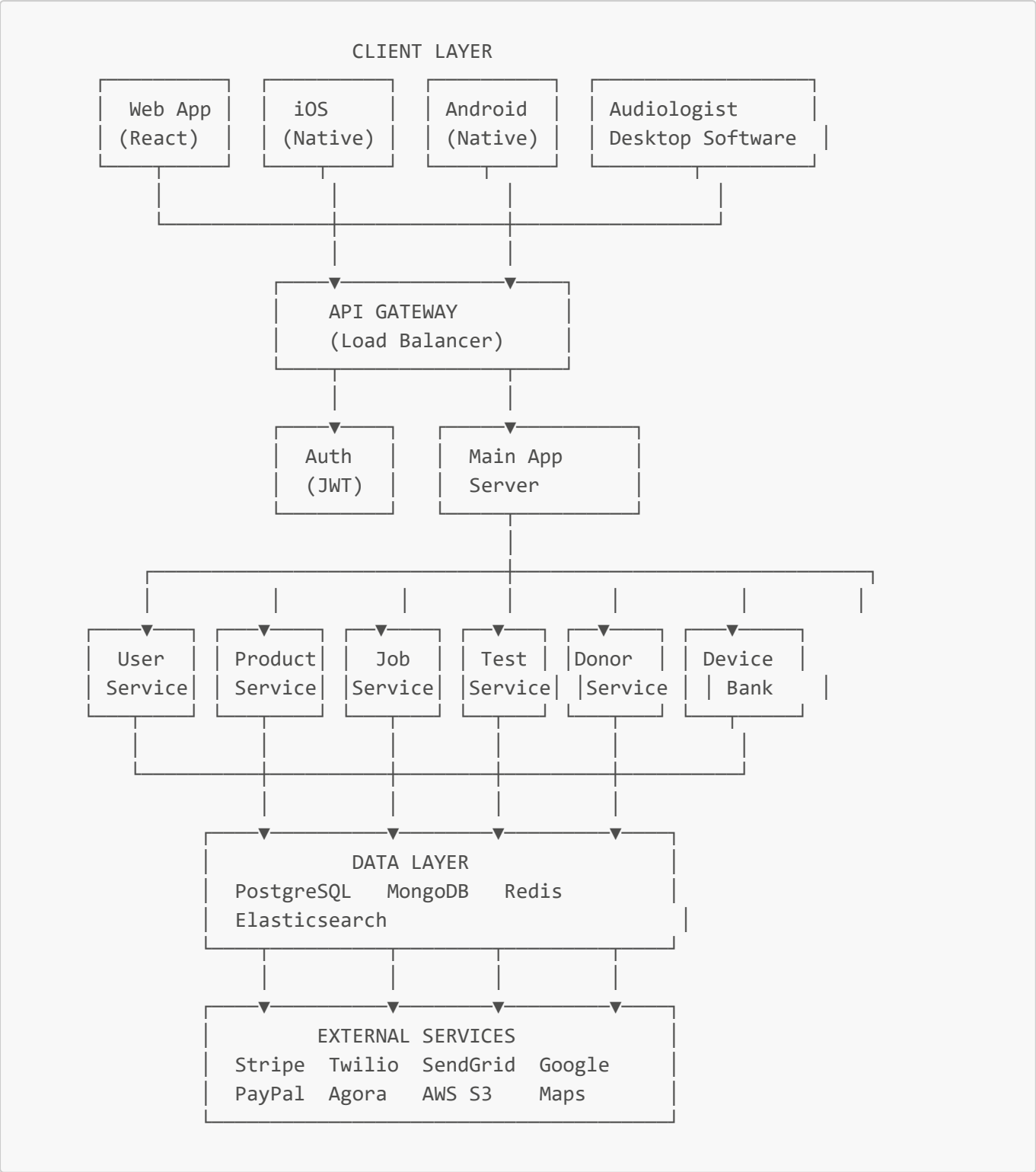
OUTPUT:

Population health dashboard + Market intelligence + Grant matching

15. Technology Stack

Layer	Recommended Technology
Frontend	React.js / Next.js (SSR for SEO)
Mobile	React Native or Flutter
Backend	Node.js (Express / NestJS) or Django
Primary Database	PostgreSQL
Document Store	MongoDB
Cache	Redis
File Storage	AWS S3 or Cloudinary
Real-Time	Socket.io
Video	WebRTC + Agora or Twilio
Full-Text Search	Elasticsearch
Maps	Google Maps API or Mapbox
Payments	Stripe (global) + JazzCash / Easypaisa (local)
SMS	Twilio
Email	SendGrid or Resend
AI / ML	TensorFlow.js or OpenAI API
CI / CD	GitHub Actions + Docker
Cloud	AWS / GCP / Azure

16. System Architecture



Appendix A — Page / Route Map

Route	Page	Access Level
/	Home / Landing	Public
/register	Registration	Public

Route	Page	Access Level
/login	Login	Public
/dashboard	User Dashboard	Authenticated
/marketplace	Product Listings	All
/marketplace/sell	Seller Dashboard	Seller
/marketplace/add-product	Add Product	Seller
/marketplace/admin-approval	Admin Approval Queue	Admin
/zakat-register	Zakat Patient Registration	Public
/donors	Donor Platform	All
/device-bank	Equipment Bank	All
/jobs	Job Portal	All
/jobs/post	Post Job	Institution
/education	Education Portal	All
/education/post	Post Programme	Institution
/scholarships	Scholarship Portal	All
/scholarships/post	Post Scholarship	Institution
/providers	Location Directory	All
/test-booking	Book Hearing Test	All
/test-dashboard	Conduct Test	Audiologist
/patient-passport	Patient Passport	Patient
/about	About Us	Public
/contact	Contact Us	Public
/faq	FAQ	Public
/community	Community Forum	Authenticated
/tinnitus	Tinnitus Support	All

Appendix B — Admin Roles & Privileges

Admin Type	Privileges
Super Admin	Full system access, user management, all approvals, settings, analytics
Content Admin	Blog, FAQ, educational content, community moderation
Marketplace Admin	Product approvals, dual-price management, seller verification

Admin Type	Privileges
Zakat / Donation Admin	Verify applications, mediate donations, manage device bank
Test Admin	Verify audiologist credentials, manage test module, quality control
Support Admin	User queries, complaints, dispute resolution

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AudSpeX — Global Hearing Healthcare Ecosystem

Complete Product Design Blueprint · v2.0 · June 2026

Every module in this document can be implemented independently as a microservice and integrated into the unified platform. This document is the intellectual property of the author and is intended for project development purposes only.